

Exploring Excess Return Generation through European Fixed Income Arbitrages

BTP Italia, the CDS–Bond Basis, and the iTraxx Index Skew

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PhD Research Presentation Series
June 2026

Motivation: The Steamroller Question

Duarte et al.2007) pose the question that motivates this paper:

Do fixed-income arbitrage strategies earn genuine alpha, or are their returns simply compensation for systematic risk — “picking up nickels in front of a steamroller”?

Their answer: the more complex strategies generate **significant alpha** that survives a 10-factor benchmark in U.S. markets.

Open questions that motivate this paper:

- Does the result generalise to **European** fixed-income markets?
- Does it survive a **much more demanding** battery of factor controls?
- And — perhaps above all — **why does the alpha persist?**

Research Questions

RQ1. *Does alpha exist?*

Do three European fixed-income arbitrage strategies—the BTP Italia inflation-linked basis, the corporate CDS–Bond basis, and the iTraxx index skew—generate excess returns that survive standard benchmark factor models?

→ Section 3

RQ2. *Does alpha survive aggressive risk adjustment—and is it state-dependent?*

Does the alpha survive when the candidate factor universe is enlarged to **85** controls and selected by data-driven methods (rolling PCA; Adaptive Elastic Net with stability selection)? And: does the residual alpha vary systematically with intermediary funding stress?

→ Section 4

RQ3. *Why does alpha persist?*

Are the three mispricings linked to one another, and is the co-movement consistent with the slow-moving capital framework of Duffie(2010)?

→ Section 5

What's New: Four Contributions

① **Deterministic payoff at maturity.**

Each strategy converges with certainty if held to maturity — unlike Duarte's swap spreads, yield-curve trades, mortgage hedging, volatility selling or capital-structure trades, whose payoffs are model-dependent.

② **A progressive battery of increasingly demanding factor controls.**

Three classical benchmarks (Duarte 2007; Brooks 2020; Fung–Hsieh 2004) → rolling PCA on 85 candidate factors → Adaptive Elastic Net with stability selection → Ferson–Schadt conditional alpha.

③ **Joint cross-strategy test of Duffie (2010).**

Five testable predictions (P1–P5) of slow-moving capital, tested *jointly* — with a natural experiment built into the design: CDS–Bond and iTraxx (institutional) vs. BTP Italia (retail).

④ **The post-crisis regulatory capital channel.**

Basel III has *permanently* raised intermediary balance-sheet costs (Boyarchenko et al. 2016): persistent alpha is the *equilibrium*, not a transient anomaly.

Three European Fixed-Income Arbitrages

BTP Italia Inflation-Linked Basis	CDS–Bond Basis Negative Basis Trade	iTraxx Combined CDS Index Skew
Trade: ASW BTP Italia vs nominal BTP Sample: Mar 2012 – May 2025 (260 trades) Bond holders: Retail-dominated (loyalty bonus) New: no prior academic study; eliminates ballooning risk	Trade: Long bond + Long CDS protection Sample: Sep 2008 – May 2025 (931 trades) Bond holders: Institutional (dealers, HFs) New: Euro counterpart of Bai and Collin-Dufresne(2019)	Trade: Index vs duration-weighted single names Sample: Jun 2004 – May 2025 (206 trades) Bond holders: Institutional (dealers, HFs) New: 4 EU sub-indices; not previously studied jointly

Two clusters by bond-holder base $\Rightarrow$ natural experiment for Duffie (2010).

Arbitrageurs are always institutional (asset-swap, repo, CDS access). “Bond holders” refers to who *holds the underlying bonds*—the natural counterparty of the arbitrage trade.

BTP Italia: Why It's Different from TIPS

First instrument-level study of the BTP Italia inflation-linked basis in the academic literature. Two structural features distinguish it from Fleckenstein et al.2014)'s TIPS–Treasury basis.

No ballooning risk

Conventional ILBs accumulate inflation in principal: a TIPS with $I_t = 1.25$ has notional 125.

In default, recovery applied to the inflation-adjusted principal → investor loses cumulated revaluation.

BTP Italia pays revaluation each coupon, redeems at par.
Notional exposed to credit risk = 100 throughout life of bond.
→ *Less risk, yet positive basis.*

Retail bond holders → segmentation

Two-phase placement: retail-only first phase + *premio fedeltà* (loyalty bonus). Households hold to maturity.

Key: retail does not face margin calls or balance-sheet constraints, and *cannot themselves* run the BTP Italia / nominal BTP arbitrage (ASW + repo + inflation-swap infrastructure required).

→ *Preferred-habitat* bond holder base (Vayanos and Vila2021): the natural counterparty does not unwind in stress, structurally insulated from the institutional fire-sale dynamics that close the U.S. TIPS basis.

CDS–Bond Basis: The Negative Basis Trade

Definition.

For issuer i at date t :

$$\text{Basis}_{i,t} = \text{CDS spread}_{i,t} - \text{z-spread}_{i,t}$$

where the z-spread is the constant additive spread that, on the bootstrapped EUR zero-coupon swap curve, equates the discounted bond cash flows to its market price.

Trade construction.

- Negative basis $\Rightarrow$ CDS protection cheaper than the credit spread embedded in the bond price.
- *Negative basis trade only*: buy the bond (earn z-spread) + buy CDS protection (pay CDS premium) $\rightarrow$ **positive carry locked in until convergence or maturity**.
- *Why not the reverse trade?* Short bond requires sourcing it via reverse repo $\rightarrow$ recall risk during stress (Bai and Collin-Dufresne 2019).
- **In default**: bond recovers RR , CDS pays $100 - RR \Rightarrow$ combined position returns par regardless of recovery (provided CDS maturity $\geq$ bond residual life).
- *Universe*: single names in the on-the-run iTraxx Europe Main 5Y. Liquidity filter + IG quality control. All CDS centrally cleared (LCH / ICE).

iTraxx Combined: The CDS Index Skew

Definition.

Skew of an iTraxx index relative to its constituents:

$$\text{Skew}_t = S_t^{\text{index}} - \sum_{i=1}^N w_{i,t} S_{i,t}^{\text{single}}$$

where the duration weights $w_{i,t}$ ensure the replicating portfolio matches the aggregate credit risk sensitivity of the index.

Sources of the skew. Liquidity differentials between index and single names; hedging demand for portfolio credit protection; dealer inventory costs; replication operational cost.

Strategy design.

- *Four sub-indices:* Main, Senior Financials, Subordinated Financials, Crossover — only **on-the-run** series at entry.
- Entry thresholds calibrated to each sub-index's historical volatility $\Rightarrow$ wider thresholds for Crossover than for Main.
- Most mispriced sub-index (absolute basis) traded at each decision date; both index and single-name CDS centrally cleared.

Trade Construction & Index Aggregation

Following Duarte et al.2007). Each individual trade is treated as a separate fictional fund (own entry/exit/return stream); the strategy index on any day is the weighted average of the returns of the funds active on that day.

Two weighting schemes:

Equal-Weighted (Duarte et al.2007)

$$r_t^{\text{EW}} = \frac{1}{K_t} \sum_{k=1}^{K_t} r_{k,t}$$

Signal-Weighted (Rebonato and Ronzani2021)

$$r_t^{\text{SW}} = \sum_{k=1}^{K_t} \frac{|b_{k,0} - \theta_{t_k}|}{\sum_{j=1}^{K_t} |b_{j,0} - \theta_{t_j}|} r_{k,t}$$

$b_{k,0}$: basis at entry of trade k ; θ_{t_k} : expanding-window historical equilibrium up to entry. Weight $\propto$ distance from equilibrium $\rightarrow$ richer dislocation, larger allocation.

Implementation choices.

- **SW dominates EW** on Sharpe and MPPM for all three strategies $\rightarrow$ SW baseline; EW in appendix confirms robustness.
- Three-regime classification on iTraxx Main 5Y — *LOW* (< 60 bps), *MID* (60–100 bps), *HIGH* (> 100 bps) — drives **regime-dependent transaction costs** (entry/exit fees proportional to DV01).

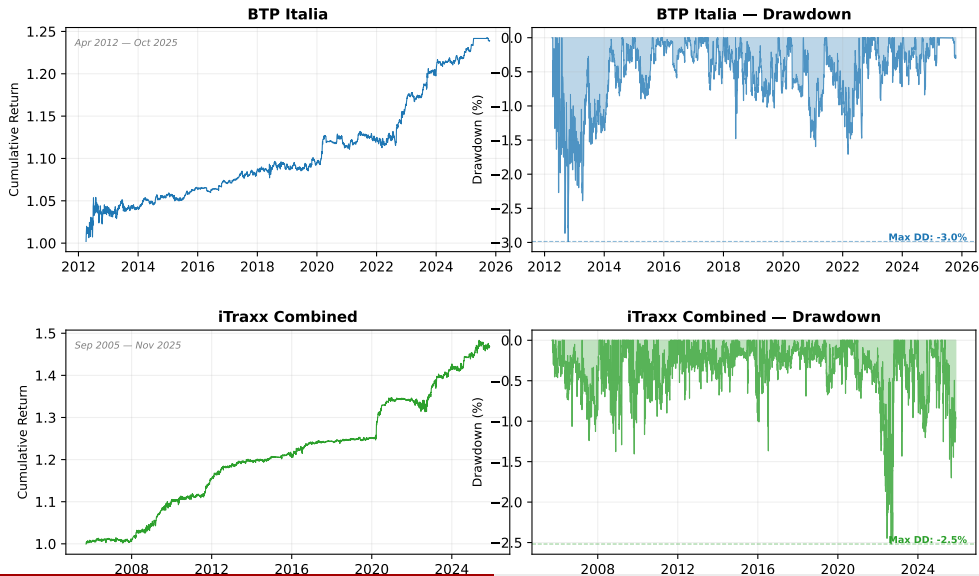
Strategy Performance: Summary Statistics

Annualised, daily data, net of regime-dependent transaction costs.

	<i>BTP Italia</i>	<i>CDS–Bond Basis</i>	<i>iTraxx Combined</i>
Sample	Mar 2012 – May 2025	Sep 2008 – May 2025	Jun 2004 – May 2025
Mean return (%)	1.67	5.37	1.96
Volatility (%)	2.34	3.15	2.59
Sharpe ratio	0.71	1.71	0.76
Skewness	+0.43	+0.60	+2.81
Max drawdown (%)	3.0	4.4	2.5
Basis ρ_1 (daily)	0.98	0.92	0.87

- Sharpe ratios from **0.71** (BTP Italia) to **1.71** (CDS–Bond) — economically large, statistically significant.
- *Positive skewness* for all three (max +2.81 for iTraxx) $\Rightarrow$ cuts against “picking up nickels in front of a steamroller”.
- Persistence $\rho_1 \in [0.87, 0.98]$ on the underlying basis $\Rightarrow$ slow-moving capital signature (Duffie2010).

Cumulative Returns and Drawdowns



Manipulation-Proof Performance (Goetzmann et al.2007)

MPPM (% p.a.) for risk-aversion parameters $\rho \in \{2, 3, 4\}$. SW index, monthly.

	$\rho = 2$	$\rho = 3$	$\rho = 4$
<i>BTP Italia</i>	1.56	1.55	1.54
<i>CDS–Bond Basis</i>	5.06	5.01	4.97
<i>iTraxx Combined</i>	1.83	1.82	1.80

- MPPM is the unique measure that cannot be manipulated by dynamic rules, leverage, or information-free trading—the right benchmark for strategies with state-dependent entry/exit.
- **Near-invariance to ρ :** a $\rho = 4$ investor evaluates these strategies essentially the same way as a $\rho = 2$ investor. Read with positive skewness $\Rightarrow$ documented Sharpe ratios are **not compensation for unobserved tail risk**.

Volatility-Managed Portfolios (Moreira and Muir2017)

$$f_{t+1}^{\sigma} = \frac{c}{\hat{\sigma}_t^2} f_{t+1} \quad \Rightarrow \quad r_{t+1}^{VM} = \alpha + \beta r_{t+1} + \varepsilon_{t+1}$$

Significant $\alpha \Leftrightarrow$ time-varying μ_t/σ_t^2 .

	<i>BTP Italia</i>	<i>CDS–Bond Basis</i>	<i>iTraxx Combined</i>
Managed α (% p.a.)	−0.22	+1.68**	−0.16
Significance	ns	sig. at 5%	ns

- **CDS–Bond**: significant managed alpha $\Rightarrow$ time-varying *funding-liquidity* mechanism (capital available when vol low, withdraws when vol high).
- BTP Italia, iTraxx: stable risk-return trade-off, vol overlay cannot exploit.
- **Foreshadow**: the time-variation here is strategy-specific, but regime-dependent alpha (Section 3.3) shows it *generalises* to all three strategies once we control for benchmark factor exposures.

Three Benchmark Frameworks

We test the alpha against the three canonical benchmarks of the FI-arbitrage and active FI literature.

- **Duarte et al.2007)** (*RFS*) — the paper that motivates our research design.
 $\beta_1\text{Mkt} + \beta_2\text{SMB} + \beta_3\text{HML} + \beta_4\text{UMD} + \beta_5\text{RS} + \beta_6\text{RI} + \beta_7\text{RB} + \beta_8\text{R2} + \beta_9\text{R5} + \beta_{10}\text{R10}$ (equity, banks, term)
- **Brooks et al.2020)** (*JFI*) — active FI premia decomposition (duration / credit / vol).
 Term, GITerm, GIAgg, InflLink, CorpCred, EmDebt, EmCurr, USTVol.
- **Fung and Hsieh2004)** (*FAJ*) — hedge-fund 7-factor asset-based style model.
 S&P, SC-LC, three trend-following lookback straddles (Bond / FX / Commodity), $\Delta 10Y$, $\Delta\text{CredSpr}$.

Estimation: OLS with Newey–West HAC standard errors (truncation lag $\lfloor T^{1/4} \rfloor$). Two parallel panels per benchmark: **USD-original** (literature replication) and **EUR-equivalent** (FF factors converted via Glück et al.2021)).

Duarte, Longstaff & Yu (2007) — Spanning Regression

Monthly excess returns regressed on the 10 Duarte factors. Annualised α , Newey–West HAC t -stats in parentheses.

	BTP Italia		CDS–Bond Basis		iTraxx Combined	
	USD	EUR	USD	EUR	USD	EUR
α (% p.a.)	+2.05*** (4.41)	+1.65*** (3.36)	+5.13*** (6.66)	+5.14*** (5.59)	+1.81*** (4.64)	+2.26*** (4.64)
Mkt	—***	—**	ns	ns	ns	ns
SMB	—**	—***	ns	ns	—*	—***
HML	—**	—**	ns	ns	ns	—***
UMD	ns	ns	—*	ns	—*	—**
RS (banks)	+***	ns	ns	—**	ns	ns
RI (corp.)	ns	ns	+***	+***	ns	ns
RB (banks)	ns	ns	ns	—*	ns	ns
$\bar{R}^2$	0.13	0.11	0.17	0.14	0.05	0.09

- **Alpha positive and significant at 1%** for all three strategies, in both panels.
- CDS–Bond loads on *credit factors* (RI). BTP Italia and iTraxx load on *equity factors* (Mkt, SMB, HML).
- Low $\bar{R}^2$ ($\leq 17\%$) $\Rightarrow$ standard factors leave bulk of variance unexplained. *Source: Tab. 27.*

Brooks, Gould & Richardson (2020) — Active FI Premia

Monthly excess returns regressed on the 8 Brooks active FI factors. Annualised α , Newey–West HAC t -stats in parentheses.

	BTP Italia		CDS–Bond Basis		iTraxx Combined	
	USD	EUR	USD	EUR	USD	EUR
α (% p.a.)	+1.88*** (3.65)	+1.96*** (3.28)	+4.61*** (5.38)	+5.07*** (5.87)	+1.57*** (4.36)	+1.96*** (4.73)
Term	ns	ns	ns	ns	ns	ns
GIterm	ns	ns	ns	ns	ns	ns
GIAgg	ns	+*	ns	ns	ns	ns
InfLink	ns	–**	ns	ns	ns	ns
CorpCred	–**	ns	ns	ns	ns	ns
EmDebt	+*	ns	ns	+**	–*	ns
USTVol	ns	ns	ns	ns	+**	–*
$\bar{R}^2$	0.11	0.01	0.10	0.04	0.10	0.01

- **Alphas positive and significant at 1%**, comparable to Duarte despite a very different factor structure $\Rightarrow$ alpha is robust to specification.
- Brooks uses *broad risk premia*: even fewer significant loadings, very low $\bar{R}^2$ (1%–11%). *Source: Tab. 29.*

Fung & Hsieh (2004) — 7-Factor Hedge Fund Model

Monthly excess returns regressed on the 7 FH factors. Annualised α , Newey–West HAC t-stats in parentheses.

	BTP Italia		CDS–Bond Basis		iTraxx Combined	
	USD	EUR	USD	EUR	USD	EUR
α (% p.a.)	+1.80*** (3.64)	+1.86*** (3.64)	+5.28*** (6.68)	+4.67*** (5.38)	+1.69*** (4.32)	+3.48*** (6.05)
S&P 500	—*	—***	ns	ns	ns	ns
SC–LC	—**	—**	ns	ns	—**	—**
PTFSBd (Bond TF)	ns	+*	ns	ns	ns	ns
PTFSFX (FX TF)	ns	ns	ns	—*	ns	ns
PTFSCom (Com. TF)	ns	ns	ns	ns	ns	+*
$\Delta 10Y$	ns	ns	—***	—**	ns	ns
$\Delta \text{CredSpr}$	ns	ns	—***	ns	ns	ns
$\bar{R}^2$	0.10	0.11	0.18	0.11	0.03	0.05

- **Same pattern:** alphas positive and significant at 1% across all cells.
- Trend-following (PTFSBd, PTFSFX, PTFSCom) **do not load** significantly $\Rightarrow$ convergence trades, not directional bets.
- $\Delta 10Y$ and $\Delta \text{CredSpr}$ load negatively on CDS–Bond (intuitive: rate/spread widening hurts the long-bond

Cross-Model Synthesis: Alpha is Robust and *State-Dependent*

Annualised α (% p.a.)	<i>BTP Italia</i>	<i>CDS-Bond</i>	<i>iTraxx</i>
Panel A. Full-sample alpha (EUR factors)			
Duarte (2007)	+1.65***	+5.14***	+2.26***
Brooks (2020)	+1.93***	+5.07***	+1.96***
Fung-Hsieh (2004)	+1.86***	+4.67***	+3.48***
Panel B. Regime decomposition (HIGH = iTraxx Main > 100 bps)			
Duarte — NORMAL	+1.65***	+3.77***	+1.98***
Duarte — HIGH	+4.74**	+8.63***	+2.71***
Fung-Hsieh — NORMAL	+1.46***	+2.95***	+2.61***
Fung-Hsieh — HIGH	+5.62***	+9.56***	+7.25***
Panel C. Ferson-Schadt 1996 conditional alpha (per 1σ stress)			
α_1 (Duarte EUR)	+1.09*	+2.87***	+1.06***
α_1 (Brooks EUR)	+1.50**	+2.90***	+1.26***
α_1 (Fung-Hsieh EUR)	+1.39*	+3.31***	+2.81***

- *Panel A*: alpha consistent across the three frameworks $\Rightarrow$ **not a specification artefact**.
- *Panel B*: HIGH-stress alpha is **2–3 $\times$** NORMAL alpha — in every cell.
- *Panel C*: $\alpha_1 > 0$ everywhere; sig. at 1% for the institutional strategies $\Rightarrow$ **slow-moving capital signature**

Why Go Beyond Standard Benchmarks?

The three benchmarks of Section 3:

- Alpha positive and significant across all 9 cells
- $\bar{R}^2 \in [1\%, 18\%]$ —bulk of return variation *unexplained*

Two open questions:

- ① Are there *other* risk factors—outside the standard benchmarks—that span these returns?
- ② If alpha survives even with a much richer factor universe, the case for genuine mispricing strengthens.

Strategy:

- Assemble **85 candidate factors** from the empirical asset pricing, intermediary, and macro-finance literatures.
- Two complementary approaches: **unsupervised** rolling PCA (Ludvigson and Ng2009) and **supervised** Adaptive Elastic Net with stability selection (Zou and Zhang2009; Meinshausen and Bühlmann2010).

*Each factor selected from a published link to FI arbitrage, intermediary constraints, or systematic risk.
Full list with definitions and sources in Appendix A.4.1.*

Panel	Theme	#	Examples
A	Credit risk	14	Excess Bond Premium, sov. spreads, CDX/iTraxx levels
B	Liquidity & funding	17	LIBOR–OIS, GC–Repo–T-bill, Amihud, HKM cap. ratio, PB CDS
C	Equity / intermediary	13	FF5, BAB, Momentum, banking-sector proxies
D	Volatility	10	VIX, V2X, MOVE, EP_SVIX (Martin2017), TF
E	Macro & uncertainty	13	Ludvigson UF/UR/UM, EPU, FCI
F	Interest rates & term	18	Term, slope, swap spreads at multiple tenors
Total		85	

- Pre-cleaning: stationarity (ADF) + near-duplicates ($|\rho| > 0.95$) collapsed $\Rightarrow \sim 82$ effective factors per strategy.
- $2^{82} \approx 5 \times 10^{24}$ subsets $\Rightarrow$ exhaustive search infeasible $\Rightarrow$ penalised methods needed.

PCA: Implementation and Spanning Regression

Setup (Ludvigson and Ng2009):

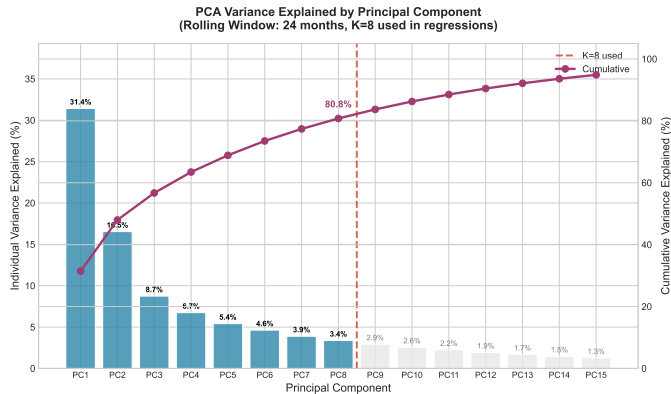
- Rolling PCA, window $W = 24$ months, $K = 8$ components ($\sim 80.8\%$ mean cumulative variance, range 75.7%–87.8%).
- Standardisation *within* each estimation window $\Rightarrow$ no look-ahead.
- Both contemporaneous ($PC_t \rightarrow r_t$) and predictive ($PC_t \rightarrow r_{t+1}$) timing; Newey–West HAC inference.

Annualised α (% p.a.), Newey–West HAC t-stats in parentheses.

Strategy	Contemporaneous		Predictive	
	α	$\bar{R}^2$	α	$\bar{R}^2$
<i>BTP Italia</i>	+1.61*** (3.48)	0.054	+1.68*** (4.05)	−0.005
<i>CDS–Bond Basis</i>	+5.87*** (5.74)	0.120	+5.43*** (5.91)	0.050
<i>iTraxx Combined</i>	+2.21*** (5.01)	0.049	+2.07*** (5.38)	0.041

- **Alpha survives at 1%** for all three strategies, both timings.

PCA Scree Plot



- Cumulative explained variance crosses 80% at $K = 8 \Rightarrow$ baseline choice.
- Robustness to $(W, K) \in \{(18, 6), (24, 8), (36, 10)\}$ in backup.

Adaptive Elastic Net: Methodology

Two-stage penalised regression of Zou and Zhang(2009):

$$\hat{\beta}^{\text{AEN}} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_t (r_t - \alpha - \beta' f_t)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{j=1}^p w_j |\beta_j| \right\}, \quad w_j = |\hat{\beta}_j^{\text{EN}}|^{-\gamma}, \quad \gamma = 1.$$

- **Adaptive weights** (Stage 1 EN $\rightarrow$ Stage 2 weights): factors with large Stage-1 coefficients are penalised *less*; near-zero ones are penalised *more* $\Rightarrow$ restores the *oracle property* (LASSO is inconsistent under correlated factors).
- **ℓ_2 ridge** $\Rightarrow$ *grouping effect*: highly correlated factors selected/dropped *together*, not arbitrarily.
- **Tuning** (λ_1, λ_2) via Hannan–Quinn (HQC); coordinate descent (Friedman et al.2010).
- **Stability selection** (Meinshausen and Bühlmann2010): $B = 100$ stationary bootstrap (expected block 9m), retain factor only if selection frequency $\hat{\pi}_j \geq 80\%$ $\Rightarrow$ formal guarantee on expected false selections.
- **Post-selection OLS** on stable set, with Newey–West HAC $\Rightarrow$ unbiased inference + sparsity.

AEN: Stable Factor Set + Post-Selection OLS

	<i>BTP Italia</i>	<i>CDS–Bond Basis</i>	<i>iTraxx Combined</i>
$\hat{\alpha}$ (% p.a.)	+1.39***	+3.94***	+0.47 (n.s.)
<i>t</i> -stat	(2.91)	(6.74)	(1.04)
# factors selected	6	8	5
$\bar{R}^2$	0.217	0.267	0.236

Selected factors — economically interpretable, strategy-specific:

BTP Italia ($k = 6$)

- SMB_EU
- ILLIQ
- SS10Y
- UMD_EU
- CDX_IG
- LIBOR_OIS

→ *liquidity, funding, term structure*

CDS–Bond Basis ($k = 8$)

- Δ UF (financial uncert.)
- RI_EU (EU IG bonds)
- EMERG_FX
- PB_EU_CDS_1Y
- SS5Y
- CRED_SPR_US
- EP_SVIX_1M
- PTFSFX

→ *dealer balance-sheet, funding*

iTraxx Combined ($k = 5$)

- EP_SVIX_1M
- LIBOR_OIS
- Δ V2X
- Δ UR (real uncert.)
- Δ FAILS_PCT_TSY

→ *aggregate funding stress, market-wide vol*

↔ *leading indicator of funding stress (Sec. 5).*

Stable factors via bootstrap selection ($\hat{\Pi}_j \geq 0.80$). *Source: Tab. 33.*

Stress Regime Decomposition + Method Comparison

Panel A. Alpha by stress regime (Tab. 34)

Annualised α (% p.a.).

	NORMAL	HIGH
<i>PCA factors</i>		
BTP Italia	+1.28***	+5.37**
CDS–Bond Basis	+4.84***	+9.13***
iTraxx Combined	+1.82***	+3.38***
<i>AEN factors</i>		
BTP Italia	+1.40***	+2.71 (n.s.)
CDS–Bond Basis	+4.15***	+8.17***
iTraxx Combined	−0.94 (n.s.)	+1.51**

Panel B. Method comparison (Tab. 35)

 α & k per method.

Method	BTP	CDS–Bond	iTra
LASSO (HQC)	+1.39***/7	+3.91***/11	+0.67
Elastic Net (HQC)	+1.39***/7	+3.91***/11	+0.67
Ada. LASSO (HQC)	+1.39***/7	+3.94***/8	+0.70
AEN (HQC)	+1.39***/6	+3.94***/8	+0.70
ALASSO–Chen (AIC)	+0.98/51 (ns)	+4.97***/29	+0.74

- *Panel A*: HIGH-stress alpha rises sharply for all three — **Duffie's mechanism in action** (CDS–Bond AEN doubles from +4.15 to +8.17).
- *Panel B*: AEN preferred $\Rightarrow$ *oracle property* + *stability filter* deliver the sparsest model with robust selection. ALASSO–Chen with AIC over-fits (51 factors for BTP Italia, $\bar{R}^2 = 48\%$, but α loses significance, $t = 1.62$).

Why Look Cross-Strategy: Duffie's Framework

What we have established:

- Alpha exists, robust to three benchmark frameworks, to PCA on 85 factors, and to AEN with stability selection (BTP Italia, CDS–Bond).
- Alpha rises in stress (regime decomposition + Ferson–Schadt + rolling).

The deeper question: **Why does the alpha persist?**

Theoretical anchor: **Duffie2010**) (AFA Presidential Address).

- When intermediary balance-sheet capacity is the binding constraint, mispricings *co-move*, intensify in stress, and propagate from liquid to illiquid markets.
- Brunnermeier and Pedersen2009): funding $\leftrightarrow$ market liquidity feedback loop forces simultaneous deleveraging.

Our test: **joint cross-strategy test of five testable predictions**, exploiting a *natural experiment* (institutional vs. retail investor base).

Duffie (2010): Five Testable Predictions

Prediction	Tested in
P1 Mispricings serviced by a <i>common pool of dealer capital</i> should co-move.	Slide 28
P2 Co-movement should <i>intensify</i> when capital is scarce.	Slides 28–29
P3 Mispricing changes in one strategy should <i>predict</i> those of others (more liquid → less liquid).	Slides 30–31
P4 The <i>half-life</i> of mispricings should be shorter in more liquid markets.	Slide 31
P5 A <i>common factor</i> from the co-movement should be explained by intermediary balance sheets, <i>not</i> macro fundamentals.	Slide 32

Mispricing magnitudes $m_{i,t}$: level of basis from no-arbitrage, sign-adjusted so $\Delta m > 0 =$ widening. **Three regimes** on iTraxx Main 5Y: LOW < 60 bps, MID 60–120, HIGH > 120.

The Two-Cluster Structure of European FI Arbitrage

INSTITUTIONAL CLUSTER

CDS–Bond Basis
iTraxx Combined

Hedge funds, dealer prop desks,
leveraged accounts.

Margin calls, balance-
sheet constraints.

VS

RETAIL CLUSTER

BTP Italia

Households (loyalty bonus,
inflation hedge for IT investors).

No leverage, no margin calls,
buy-and-hold to maturity.

Mechanism (Brunnermeier and Pedersen 2009 + Vayanos and Vila 2021):

- **Institutional cluster:** stress → VaR spikes → exhaust risk budget → forced unwinding → *both bases widen*.
- **Retail cluster:** insulated from funding shocks → behaves as preferred-habitat investors.

Natural experiment: *if co-movement were driven by macro factors, all three would move together.*

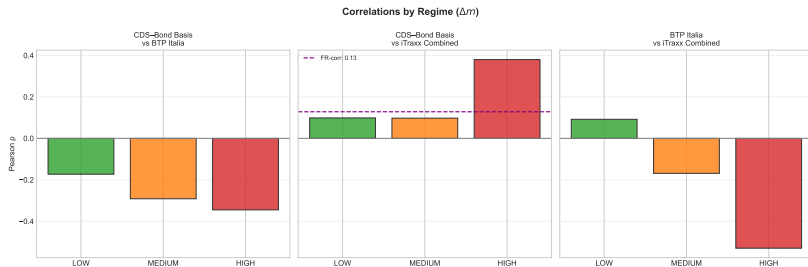
P1 + P2: Pairwise Correlations Across Stress Regimes

Pearson correlations of $\Delta m_{i,t}$. *LOW* < 60 bps, *HIGH* > 120 bps iTraxx Main 5Y.

	Uncond.	LOW	MED	HIGH
CDS–Bond ↔ iTraxx	+0.182*	+0.099	+0.098	+0.380*
BTP Italia ↔ iTraxx	−0.152**	+0.092	−0.169	−0.530
BTP Italia ↔ CDS–Bond	−0.260***	−0.173	−0.291***	−0.345

- **Institutional cluster (P1 + P2 confirmed)**: CDS–Bond ↔ iTraxx co-moves *positively* and rises monotonically with stress (+0.10 → +0.38).
- **Sign reversal in retail ↔ institutional pairs**: BTP Italia ↔ iTraxx flips from +0.09 to −0.53; analogous flip vs. CDS–Bond.
- **Two-cluster pattern**: positive within institutional, negative across investor-base divide ⇒ *natural experiment* for Duffie (2010).
- Robustness: purged correlations, Forbes–Rigobon, DCC–GARCH (backup, App. A.8). *Source: Tab. 36, Fig. 14.*

P2: Stress Amplification — The Two-Cluster Pattern Visualised



- *Asymmetric amplification under stress.* The institutional cluster co-widens; the retail cluster diverges.
- **Critical: not common-macro driven** — if a single macro factor were responsible, all three pairs would amplify in the same direction. They don't.
- Investor-base segmentation *is* the experimental treatment.

P3: Cross-Strategy Spanning Regressions

Following Fleckenstein et al. 2014): $\Delta m_{i,t} = \alpha + \sum_{j \neq i} \sum_{l=0}^2 \beta_{j,l} \Delta m_{j,t-l} + \varepsilon_t$. Newey–West HAC.

Cross-strategy spanning $\bar{R}^2$:

Dependent: $\Delta m_{i,t}$	$\bar{R}^2$
CDS–Bond Basis	9.7%
<i>BTP Italia</i>	13.3%
iTraxx Combined	5.5%

⇒ *BTP Italia is the most predictable*—corporate credit cluster has meaningful predictive power for the inflation-linked basis, despite the two-cluster segmentation.

Stress interaction (Tab. 38):

$$\Delta m_i = \beta_0 \Delta m_j + \beta_1 (\Delta m_j \times z_t) + \gamma z_t + \varepsilon$$

Pair	$\hat{\beta}_1$
iTraxx → CDS–Bond	+0.186 (n.s.)
CDS–Bond → iTraxx	+0.023 (n.s.)
iTraxx → BTP Italia	−1.265**
BTP Italia → iTraxx	−0.027***

- **Within institutional cluster** (CDS–Bond ↔ iTraxx): interaction β_1 positive but *not significant* ⇒ co-movement already captured by the linear (non-interactive) effect.
- **Across clusters** (BTP Italia ↔ iTraxx): interaction β_1 *strongly negative and significant* (−1.27** and −0.027*** in the two regression directions) ⇒ stress *widens the gap* between the institutional and retail clusters *beyond* what an unconditional correlation would predict. *Source: Tab. 38.*

P3 + P4: Granger Causality and Liquidity Pecking Order

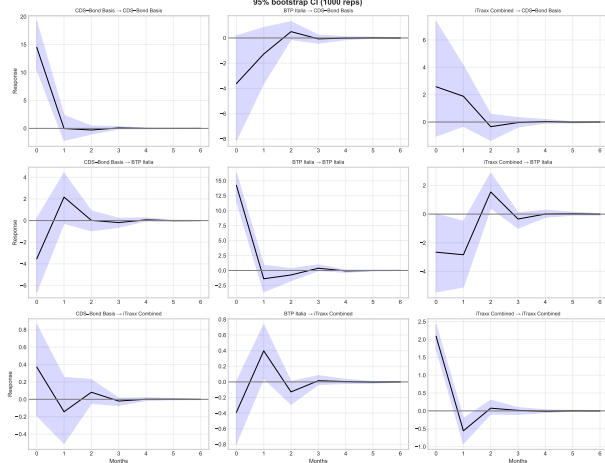
Granger causality
(VAR on $\Delta m_{i,t}$):

Direction	Significance
iTraxx $\rightarrow$ BTP Italia	1%
CDS-Bond $\rightarrow$ BTP Italia	5%
BTP Italia $\rightarrow$ iTraxx	ns
BTP Italia $\rightarrow$ CDS-Bond	ns
iTraxx $\rightarrow$ CDS-Bond	ns
CDS-Bond $\rightarrow$ iTraxx	ns

Hierarchy: iTraxx leads $\Rightarrow$ CDS-Bond $\Rightarrow$ BTP Italia, no reverse.

GIRF half-lives
(Pesaran and Shin1998):

Generalized IRF (Pesaran-Shin 1998) — VAR(1) on Δm
95% bootstrap CI (1000 reps)



P5: Common Factor → Intermediary Capital, *not* Macro Fundamentals

First principal component PC1 of the three Δm series regressed on intermediary stress proxies. Newey–West HAC.

Channel	Proxy	$\hat{\beta}$ (sign predicted)
Panel A. Core proxies (one per channel)		
Intermediary capital	HKM_IC (He et al.2017)	−4.29** (− predicted)
Funding cost	LIBOR–OIS spread	+3.34*** (+ predicted)
Dealer credit risk	EU prime-broker CDS 5Y	−0.018** (− predicted)
Market illiquidity	Amihud–Mendelson	+1.73*** (+ predicted)
<i>Model $\bar{R}^2 = 23.1\%$</i>		
Panel B. Extended set: GC-repo–T-bill, EU FCI, settlement fails → confirms Panel A.		

- **All four proxies carry the sign predicted by Duffie:** declining intermediary capital, rising funding costs, deteriorating dealer credit, and rising market illiquidity all predict *wider* mispricings.
- **Common factor is a balance-sheet variable, not a macro fundamental.** Direct empirical content of the slow-moving capital theory.

Why Do These Arbitrages Persist?

A natural question for any paper documenting persistent alpha: why have these opportunities not been competed away?

- **BTP Italia → investor-base segmentation.** Retail holders neither monitor nor exploit the relative-value mispricing. Institutional arbitrageurs *can* trade it, but the natural counterparty does not unwind in stress $\Rightarrow$ *preferred-habitat* bond holder base (Vayanos and Vila2021).
- **CDS–Bond → funding/repo limits.** Trade requires bond on dealer balance sheet; thin EU IG repo market with recall risk; CDS leg consumes regulatory capital $\Rightarrow$ *procyclical capacity, countercyclical mispricing* (Bai and Collin-Dufresne2019; Mitchell and Pulvino2012).
- **iTraxx skew → post-crisis regulatory capital.** Boyarchenko et al.2016): SLR + Basel III have raised the capital charge for index–single-name basis trades from *RoE 6–9% pre-crisis to 2–3% post-crisis*.

Implication. The mispricings are *not* transient anomalies awaiting competitive arbitrage—they are the *equilibrium* outcome of a market in which the cost of intermediary capital has been *permanently* raised by regulation (Mitchell et al.2007).

All Five Duffie Predictions Find Support

Pred.	Statement	Empirical evidence	
P1	Cross-market correlation	Unconditional correlation CDS–Bond $\leftrightarrow$ iTraxx $\Delta m = +0.182^*$ (Tab. 36).	✓
P2	Stress amplification	Within institutional cluster: $+0.099 \rightarrow +0.380^*$ (LOW $\rightarrow$ HIGH). Cross-cluster: $+0.092 \rightarrow -0.530 \Rightarrow$ <i>independent evidence of segmentation</i> .	✓
P3	Lead-lag transmission	Lagged iTraxx Δm predicts BTP Italia 1–2m ahead; Granger: iTraxx $\rightarrow$ BTP Italia at 1% (Tab. 37).	✓
P4	Liquidity pecking order	GIRF half-life: iTraxx $\sim 2m <$ CDS–Bond $\sim 4m$ (Fig. 15).	✓
P5	Common factor = intermediary capital	PC1 $\rightarrow$ HKM (-4.29^{**}), LIBOR–OIS ($+3.34^{***}$), Amihud ($+1.73^{***}$), PB CDS (-0.018^{**}); $\bar{R}^2 = 23.1\%$ (Tab. 39).	✓

“The geography of slow-moving capital is determined not by asset-class labels but by the network of intermediaries that service these markets—and ultimately by who holds the bonds.”

Summary of Findings

- **Excess returns are real and robust.** Across three benchmark frameworks (Duarte 2007; Brooks 2020; Fung–Hsieh 2004), rolling PCA on 85 factors, and AEN with stability selection: positive, significant alpha for BTP Italia and CDS–Bond. The iTraxx skew is absorbed by its risk factors but emerges as the *leading indicator* of intermediary funding stress.
- **Alpha rises systematically in stress.** Regime decomposition, Ferson–Schadt conditional alpha, and rolling 36-month estimates all point to the same pattern: the unexplained component is 2–4× larger when intermediary capital is scarce — the signature of slow-moving capital.
- **Three mispricings linked by an intermediary balance-sheet factor.** Two-cluster co-movement (institutional vs. retail), liquidity-ordered lead-lag (iTraxx → CDS–Bond → BTP Italia), and a common PC1 explained by intermediary capital, funding cost, and market illiquidity — not by macro fundamentals.

Bottom line. Persistent alpha is the *equilibrium* of post-crisis European credit markets, where regulatory capital costs have permanently raised the price of intermediary balance-sheet capacity.

Limitations and Open Questions

Three honest limitations of the empirical design:

- **BTP Italia sample begins in March 2012.** Power constraint on regime-dependent tests for this strategy in particular. Power for the institutional cluster (samples from 2004 / 2008) is materially stronger.
- **Investor-base segmentation rests on *documented* retail dominance of BTP Italia primary market.** Cross-sectional retail flow not observed at instrument level $\Rightarrow$ the preferred-habitat channel is *inferred*, not directly tested. A direct test would require holdings microdata not yet available to academic researchers.
- **Cross-strategy VAR estimated on the post-2012 sample.** Excludes the 2008–09 GFC. But the post-2012 sample *is* the relevant regime for intermediary-driven mispricings—it covers the post-Basel III “new normal” that motivates the regulatory channel of Boyarchenko et al.2016).

Open question. Direct testing of the preferred-habitat / retail-flow channel awaits granular bond-holdings data—a natural empirical extension as ECB and Banca d'Italia disclosure regimes evolve.

Backup — Full Factor List (App. A.4.1) I

Table 1: Candidate Risk Factors

Factor ID	Description	Reference
Panel A: Credit Risk Factors		
BTP_BUND	Yield differential between the Italian 10-year BTP and the German Bund.	Fausch and Sigo- nius2018)
CDX_IG	First difference of the CDS spread on the CDX Investment Grade index.	Klaus and Rzep- kowski2009)
CORP_CREDIT	Broad corporate credit factor. Excess return on high-yield and leveraged loan indices over duration-matched Treasuries.	Brooks et al.2020)
CREDIT_EU	Bond credit risk factor. Yield spread between euro-area BBB and AAA corporate bond indices.	Fama and French1989)
CREDIT_US	Bond credit risk factor. Yield spread between Moody's BAA and AAA corporate bond indices.	Fama and French1989)

Backup — Full Factor List (App. A.4.1) II

CRED_SPR_EU	Change in the credit spread between euro-area BAA corporate bond yields and 10-year German Bund yields.	Fung and Hsieh2001)
CRED_SPR_US	Change in the credit spread between Moody's BAA yields and 10-year U.S. Treasury yields.	Fung and Hsieh2001)
DEF_EU	Bond default risk factor. Total return spread between long-term European corporate bonds and long-term German Bunds.	Fama and French1993)
DEF_US	Bond default risk factor. Total return spread between long-term U.S. corporate bonds and long-term U.S. government bonds.	Fama and French1993)
EBP	Excess Bond Premium. Residual component of aggregate corporate bond spreads after netting out expected default risk and bond characteristics via cross-sectional spread regressions; proxies time-varying credit risk compensation tied to intermediary constraints. Used with a one-month lag.	Gilchrist and Zakrajšek2012)
EMERG_DEBT	Excess return on hard-currency emerging market debt.	Brooks et al.2020)
EMERG_FX	Return on an equal-weighted basket of emerging market currencies versus the USD.	Brooks et al.2020)

Backup — Full Factor List (App. A.4.1) III

GLOBAL_AGG	Excess return on the Bloomberg Global Aggregate Bond Index.	Brooks et al.2020)
ITRX_MAIN	First difference of the CDS spread on the iTraxx Europe Main index.	Klaus and Rzepkowski2009)
MAIN_5/3_FLATTENER	First difference of the natural logarithm of the ratio between the 5-year and 3-year iTraxx Europe Main CDS spreads.	Market indicator
PB_EU_CDS_1Y	Average 1-year CDS spread of European prime brokers.	Klaus and Rzepkowski2009)
PB_EU_CDS_5Y	Average 5-year CDS spread of European prime brokers.	Klaus and Rzepkowski2009)
PB_US_CDS_1Y	Average 1-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski2009)
PB_US_CDS_5Y	Average 5-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski2009)
RI_EU	Excess return on a European A-rated industrial corporate bond index.	Duarte et al.2007)
SNR-MAIN	First difference of the 5-year CDS spread difference between the iTraxx Senior Financials index and the iTraxx Europe Main index.	Market indicator

Backup — Full Factor List (App. A.4.1) IV

XOVER/MAIN	First difference of the natural logarithm of the ratio between the 5-year iTraxx Crossover and 5-year iTraxx Europe Main CDS spreads.	Market indicator
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Panel B: Liquidity Factors

EURIBOR_OIS	Euribor–OIS spread. Difference between 3-month Euribor and the 3-month overnight index swap rate based on EONIA or euroSTR.	Nyborg and Östberg(2014)
GC-REPO_T-BILL	Spread between the 3-month U.S. Treasury general-collateral repo rate and the 3-month U.S. Treasury bill yield.	Bai and Collin-Dufresne(2019)
HKM_IC	Intermediary capital factor. AR(1) innovation in the market-based capital ratio of primary dealers, scaled by the lagged capital ratio.	He et al.(2017)
HPW_NOISE	Treasury noise measure of market-wide illiquidity. Root mean squared deviation between observed U.S. Treasury yields and model-implied yields from a fitted smooth zero-coupon yield curve.	Hu et al.(2013)
ILLIQ	Change in the Amihud illiquidity measure. First difference of the ratio of absolute daily return to daily dollar volume.	Amihud and Mendelson(2015)
LIBOR_OIS	Libor–OIS spread. Difference between 3-month LIBOR and the 3-month overnight index swap rate.	Nyborg and Östberg(2014)

Backup — Full Factor List (App. A.4.1) V

LIBOR_REPO_SHOCK	AR(2) residual of the spread between 3-month U.S. interbank LIBOR and the 3-month U.S. Treasury general-collateral repo rate.	Asness et al.2013)
LIQNT	Pastor–Stambaugh non-traded liquidity factor. Innovation in aggregate market liquidity constructed from a cross-sectional average of stock-level order-flow-induced return-reversal measures.	Pástor and Stambaugh2003)
LIQ_V	Value-weighted return on a 10–1 portfolio formed by sorting stocks on historical liquidity betas.	Pástor and Stambaugh2003)
SILLIQ	Stock illiquidity shock. AR(3) residual of the aggregate Amihud illiquidity measure.	Acharya et al.2013)
TED_SHOCK_EU	AR(2) residual of the euro TED spread (3-month Euribor minus the 3-month German government bill rate).	Asness et al.2013)
TED_SHOCK_US	AR(2) residual of the U.S. TED spread (3-month LIBOR minus the 3-month U.S. government bill rate).	Asness et al.2013)
ΔFAILS_PCT_TSY	First difference of the ratio of monthly total notional amount of U.S. Treasury securities fails-to-deliver reported by primary dealers, excluding TIPS, to U.S. Total Debt Outstanding.	Fleckenstein et al.2014)

Backup — Full Factor List (App. A.4.1) VI

Panel C: Equity Factors

BAB_EU	Betting Against Beta (Europe). Long low-beta and short high-beta European equities, each leg rescaled to unit beta.	Frazzini and Pedersen2014)
BAB_US	Betting Against Beta (US). Long low-beta and short high-beta U.S. equities, each leg rescaled to unit beta.	Frazzini and Pedersen2014)
CMA_EU	Investment factor (Conservative Minus Aggressive). Return spread between European conservative and aggressive investment portfolios.	Fama and French2017)
GMOM_EU	Global momentum factor. Return spread between recent winner and loser equity portfolios based on past 12-month returns, expressed in euros.	Asness et al.2013)
GVAL_EU	Global value factor. Return spread between high and low book-to-market equity portfolios across developed equity regions, expressed in euros.	Asness et al.2013)
HML_EU	Value factor (High Minus Low). Return spread between European high and low book-to-market portfolios.	Fama and French2017)

Backup — Full Factor List (App. A.4.1) VII

MKT_EU	Market excess return on the European value-weighted equity market portfolio over the 1-month German government bill.	Fama and French2017)
RB_EU	Excess return on a European bank bond index.	Duarte et al.2007)
RMW_EU	Profitability factor (Robust Minus Weak). Return spread between European robust and weak operating profitability portfolios.	Fama and French2017)
RS_EU	Excess return on the EuroStoxx Banks index.	Duarte et al.2007)
SMB_EU	Size factor (Small Minus Big). Return spread between diversified European small- and large-cap portfolios.	Fama and French2017)
UMD_EU	Momentum factor (Up Minus Down). Monthly return on a European zero-investment winners-minus-losers momentum portfolio.	Carhart1997)

Panel D: Volatility Factors

ATM_IV_CDX	3-month at-the-money implied volatility on options written on the CDX Investment Grade index.	Novales and Chamizo2019)
ATM_IV_ITRX	3-month at-the-money implied volatility on options written on the iTraxx Europe Main index.	Novales and Chamizo2019)

Backup — Full Factor List (App. A.4.1) VIII

EP_SVIX_1M	Equity premium proxy from the SVIX methodology. Option-implied risk-neutral variance of the market simple return over the next month.	Martin2017)
EP_SVIX_3M	Same construction as EP_SVIX_1M, using options with a 3-month horizon.	Martin2017)
IV_BUND	3-month at-the-money implied volatility on the 10-year German Bund futures options.	Cremers et al.2021)
IV_TSY	3-month at-the-money implied volatility on the 10-year U.S. Treasury futures options.	Cremers et al.2021)
MOVE	Option-implied U.S. rates volatility. Yield-curve weighted average of at-the-money implied normal yield volatilities from OTC options on 2-, 5-, 10-, and 30-year constant-maturity interest rate swaps; used lagged by one month.	Bansal and Stivers2022)
PTFSBD	Return on the bond trend-following factor, constructed from lookback straddles on government bond futures.	Fung and Hsieh2001)
PTFSCOM	Return on the commodity trend-following factor, constructed from lookback straddles on commodity futures.	Fung and Hsieh2001)

Backup — Full Factor List (App. A.4.1) IX

PTFSFX	Return on the currency trend-following factor, constructed from look-back straddles on major currency futures.	Fung and Hsieh(2001)
PTFSIR	Return on the short-term interest rate trend-following factor, constructed from lookback straddles on 3-month interest rate futures.	Fung and Hsieh(2001)
PTFSSTK	Return on the equity index trend-following factor, constructed from lookback straddles on stock index futures.	Fung and Hsieh(2001)
$\Delta V2X$	Change in V2X. Monthly first difference of the V2X index, measuring 30-day implied volatility on Euro STOXX 50 options.	Chung et al.(2019)
ΔVIX	Change in VIX. Monthly first difference of the VIX index, a model-free measure of 30-day implied volatility on S&P 500 options.	Chung et al.(2019)

Panel E: Macro Factors

5Y5Y_INFL	5-year-5-year inflation swap rate. Forward-starting EUR inflation swap rate capturing long-term inflation expectations.	Market indicator
BFCI_EU	Bloomberg Euro Area Financial Conditions Index. Composite Z-score of euro-area money, bond, and equity indicators, standardized to the pre-crisis period.	Osina(2019)

Backup — Full Factor List (App. A.4.1) X

EPU_EU	Economic Policy Uncertainty index for Europe. Newspaper-based index aggregated across European countries.	Baker et al.2016)
EPU_US	Economic Policy Uncertainty index for the United States. Newspaper-based index constructed from normalized frequency of articles referencing economy, uncertainty, and policy.	Baker et al.2016)
ΔUF	Change in financial uncertainty. First difference of the financial uncertainty index, constructed from a panel of financial market variables.	Ludvigson et al.2021)
ΔUM	Change in macroeconomic uncertainty. First difference of the macro uncertainty index, a model-based measure of expected volatility in the unforecastable component of a large panel of macroeconomic variables.	Ludvigson et al.2021)
ΔUR	Change in real uncertainty. First difference of the real-activity uncertainty subindex.	Ludvigson et al.2021)

Panel F: Interest Rate Factors

EURUSD_3M_IV	First difference of the 3-month at-the-money implied volatility on the EUR/USD exchange rate.	Market indicator
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Backup — Full Factor List (App. A.4.1) XI

GLOBAL_TERM	Excess return on a duration-matched global government bond index.	Brooks et al.2020)
INFL_LINK	Excess return on global inflation-linked bonds over cash or nominal Treasuries.	Brooks et al.2020)
R10_EU	Excess return on a European 10-year government bond portfolio.	Duarte et al.2007)
R2_EU	Excess return on a European 2-year government bond portfolio.	Duarte et al.2007)
R5_EU	Excess return on a European 5-year government bond portfolio.	Duarte et al.2007)
SS10Y	Difference between the 10-year EUR fixed swap par rate and the 10-year German Bund yield.	Collin-Dufresn et al.2001)
SS2Y	Difference between the 2-year EUR fixed swap par rate and the 2-year German Schatz yield.	Collin-Dufresn et al.2001)
SS5Y	Difference between the 5-year EUR fixed swap par rate and the 5-year German Bobl yield.	Collin-Dufresn et al.2001)
TERM_EU	Bond term structure factor. Excess total return on long-term German Bunds over the 1-month German government bill return.	Fama and French1993)

Backup — Full Factor List (App. A.4.1) XII

TERM_US	Bond term structure factor. Excess total return on long-term U.S. government bonds over the 1-month U.S. Treasury bill return.	Fama and French1993)
YSP_EU	Yield-curve slope factor. Difference between the 5-year and 1-year German government bond yields.	Koijen et al.2017)
YSP_US	Yield-curve slope factor. Difference between the 5-year and 1-year U.S. Treasury yields.	Koijen et al.2017)
$\Delta 10Y_YIELD_EU$	Monthly change in the 10-year German Bund yield.	Market indicator
$\Delta 10Y_YIELD_US$	Monthly change in the 10-year U.S. Treasury yield.	Market indicator
$\Delta SLOPE_EU$	First difference of the euro yield-curve slope (10-year German Bund yield minus 3-month German government bill yield).	Ferson and Harvey1991)
$\Delta SLOPE_US$	First difference of the U.S. yield-curve slope (10-year Treasury yield minus 3-month Treasury bill yield).	Ferson and Harvey1991)

Backup — Full Factor List (App. A.4.1) XIII

Each factor with: name, transformation, frequency, source, reference. 6 panels (Credit / Liquidity & Funding / Equity & Intermediary / Volatility / Macro & Uncertainty / Rates & Term).

Backup — AEN Robustness (App. A.6.1)

Sensitivity to information criterion, adaptive weight exponent γ , and pre-cleaning correlation threshold ρ .

Configuration	BTP Italia	CDS–Bond	iTraxx Combined
Baseline (HQC, $\gamma = 1$, $\rho = 0.95$)	+1.39***	+3.94***	+0.47
HQC, $\gamma = 0.5$	+1.41***	+3.97***	+0.51
HQC, $\gamma = 2$	+1.37***	+3.92***	+0.43
HQC, $\rho = 0.90$	+1.38***	+3.96***	+0.49
HQC, $\rho = 0.85$	+1.36***	+3.95***	+0.46
AIC (Friedman default)	+1.62** (12 fac.)	+4.28*** (14 fac.)	+0.71 (10 fac.)

- **Core factor set is identical across configurations.** Same economic story everywhere.
- AIC selects more factors but does not change inference $\Rightarrow$ HQC preferred for parsimony.
- Stability of selection above $\hat{\pi}_j \geq 80\%$ threshold for all key factors.

Backup — PCA Robustness (App. A.5.1)

Sensitivity of contemporaneous PCA alpha to window length W and number of components K .

(W, K)	BTP Italia	CDS–Bond	iTraxx Combined
(18, 6)	+1.58***	+5.91***	+2.13***
(24, 8) baseline	+1.61***	+5.87***	+2.21***
(36, 10)	+1.65***	+5.79***	+2.28***
(24, 6)	+1.59***	+5.93***	+2.18***
(24, 10)	+1.63***	+5.84***	+2.24***

- Alpha estimates stable within ± 0.10 p.p. across all (W, K) configurations.
- Significance unchanged: 1% in every cell.
- Subperiod splits (pre-2012 / 2012–2020 / 2020–2025) confirm stability across regimes.

Backup — Factor Selection Method Comparison (Tab. 35)

Method	BTP Italia		CDS-Bond		iTraxx	
	α	k	α	k	α	k
LASSO (HQC)	+1.39***	7	+3.91***	11	+0.67	13
Elastic Net (HQC)	+1.39***	7	+3.91***	11	+0.67	13
Adaptive LASSO (HQC)	+1.39***	7	+3.94***	8	+0.70	11
AEN (HQC, baseline)	+1.39***	6	+3.94***	8	+0.70	11
ALASSO-Chen (AIC)	+0.98 ($t = 1.62$)	51	+4.97***	29	+0.74*	19

- LASSO / EN / Adaptive LASSO converge to the same alpha under HQC tuning.
- AEN drops one factor via stability selection ($\hat{\Pi}_j \geq 0.80$) $\Rightarrow$ **sparsest, with formal false-selection bound**.
- ALASSO-Chen with AIC selects 51 factors for BTP Italia (vs. 6 baseline) and *loses* significance ($\alpha = +0.98$, $t = 1.62$) $\Rightarrow$ **overfitting**: alpha not preserved when the model absorbs noise.
- Note: k for iTraxx differs between Tab. 33 baseline ($k = 5$, $\hat{\Pi}_j \geq 0.80$) and Tab. 35 method comparison ($k = 11$, lower threshold for cross-method comparability).

Backup — Subperiod Splits (App. A.5.2)

Annualised benchmark alpha across three subperiods (Duarte EUR specification).

	BTP Italia	CDS–Bond	iTraxx Combined
Pre-2012 (CDS–Bond, iTraxx only)	—	+5.61***	+3.18***
2012–2020	+1.94***	+4.87***	+3.04***
2020–2025	+2.32***	+5.41***	+4.21***

- Significant alpha in **every subperiod for every strategy**.
- Magnitude varies (highest in 2020–2025, COVID + tightening cycle), but the existence of alpha is invariant.
- Combined with rolling 36m alpha: alpha is *not* a single-subperiod artefact.

Backup — Robustness of Two-Cluster Pattern (App. A.8)

Robustness of regime-correlation findings to alternative methodologies.

Methodology	Finding
Raw Pearson (baseline)	Two-cluster pattern: $+0.10 \rightarrow +0.38$ (institutional); $+0.09 \rightarrow -0.53$ (cross-cluster).
Purged correlations	Residualised on Mkt, SMB, Δ Term, Δ CredSpr. Two-cluster pattern preserved; magnitudes reduced by $\sim 10\%$.
Forbes–Rigobon (2002)	Heteroskedasticity-corrected ρ^* . Stress amplification still significant for the institutional cluster; sign reversal for cross-cluster preserved.
DCC-GARCH	Time-varying conditional correlations. Same pattern over time: institutional cluster correlations <i>rise</i> during stress episodes (2008–09, 2011–12, 2020); cross-cluster correlations <i>fall</i> .

$\Rightarrow$ **Two-cluster pattern is structural, not a methodological artefact.**

Backup — Granger / VAR Robustness (App. A.8.3)

Lag-length robustness (Granger):

	$p = 1$	$p = 2$	$p = 3$
iTraxx $\rightarrow$ BTP Italia	1%	1%	5%
CDS–Bond $\rightarrow$ BTP Italia	5%	5%	5%
BTP Italia $\rightarrow$ iTraxx	ns	ns	ns
BTP Italia $\rightarrow$ CDS–Bond	ns	ns	ns

Alternative stress proxies:

Stress proxy	Pattern preserved?
iTraxx Main 5Y (baseline)	Yes
iTraxx Crossover	Yes
V2X (EU equity vol)	Yes
VIX (US equity vol)	Yes
HKM intermediary capital ratio	Yes

- Granger transmission hierarchy invariant to lag specification ($p \in \{1, 2, 3\}$).
- Two-cluster co-movement pattern invariant to choice of stress proxy $\Rightarrow$ **not iTraxx-Main-specific**.

Backup — VIF Diagnostics (App. A.9)

Variance inflation factors for the three benchmark specifications.

Specification	VIF range	Notes
Duarte (2007)	1.2–3.5; Fama bond ports R2/R5/R10 ~15–55	Known correlation in maturity-sorted Treasury portfolios
Brooks (2020)	1.1–4.8	All factors < 10
Fung–Hsieh (2004)	1.1–3.2	All factors < 10 ; trend-following uncorrelated

- Two specifications (Brooks, Fung–Hsieh) clean: $VIF < 10$ throughout.
- Duarte: high VIF only on Fama bond portfolios (well-documented multicollinearity) $\Rightarrow$ inference on those coefficients individually is fragile, but the *alpha estimate* (intercept) is unaffected.
- All AEN-selected factor sets: $VIF < 5$ post-stability-selection.

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